

MWANZA, Tanzania, United Republic ofWMO: **637560**Lat: **2.444S**Lon: **32.933E**Elev: **3763**StdP: **12.80**Time Zone: **3.00 (E03)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	61.0	62.5	46.3	53.4	74.0	48.3	57.6	73.4	23.1	75.6	22.2	75.0	5.0	120	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	15.3	87.3	66.9	86.0	67.0	84.6	67.2	72.8	79.8	72.0	79.4	71.6	79.1	12.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	128.9	76.1	70.0	127.1	75.8	69.5	125.0	75.5	39.2	79.9	38.4	79.4	38.0	79.3	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	19.9	18.1	DB	55.3	90.2	3.6	1.2	52.7	91.0	50.6	91.8	48.6	92.4	46.1	93.3	(4)
(5)			WB	51.1	76.8	2.1	1.5	49.6	77.8	48.4	78.7	47.2	79.5	45.6	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.0	74.1	74.7	74.8	74.5	74.4	73.5	72.8	73.8	74.3	74.8	73.7	73.1	(6)	
	DBStd	2.11	2.30	2.38	2.41	2.03	1.54	1.78	1.70	1.63	1.95	2.04	2.04	2.10	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	8773	746	693	768	735	758	705	706	737	730	768	711	716	(10)	
	CDD65	3298	281	273	303	285	293	255	241	272	280	303	261	251	(11)	
	CDH74	20418	1690	1748	1922	1635	1818	1782	1763	1896	1824	1853	1317	1170	(12)	
	CDH80	3917	322	392	457	325	313	352	304	373	362	381	186	150	(13)	
(14) Wind	WSAvg	7.6	7.2	7.1	7.2	7.2	7.3	7.7	8.4	8.3	8.6	8.6	7.4	6.7	(14)	
(15) Precipitation	PrecAvg	39.7	3.6	3.9	5.1	7.0	3.4	0.4	0.1	0.7	1.0	3.1	6.1	5.3	(15)	
	PrecMax	51.2	7.2	9.4	13.1	13.0	9.8	2.6	1.1	3.1	2.7	7.3	9.2	9.9	(16)	
	PrecMin	26.5	1.1	0.7	0.6	1.5	0.2	0.0	0.0	0.0	0.0	0.2	1.7	1.9	(17)	
	PrecStd	7.2	1.9	2.2	3.5	2.8	2.2	0.7	0.2	0.9	0.7	2.1	2.1	2.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.0	88.2	88.7	86.7	85.7	86.3	86.3	86.7	86.8	87.7	85.9	85.3	(19)	
		MCWB	66.7	67.9	67.8	69.0	68.0	65.7	63.5	64.8	65.8	66.4	67.9	67.4	(20)	
	2%	DB	84.6	86.2	86.2	84.6	84.2	84.6	84.5	84.9	84.7	85.4	84.0	83.2	(21)	
		MCWB	68.1	67.9	68.6	69.2	68.4	65.5	63.9	64.9	65.9	67.0	68.4	68.5	(22)	
	5%	DB	83.4	84.4	84.4	83.6	82.7	83.8	82.8	83.8	83.9	84.1	82.4	81.8	(23)	
		MCWB	68.6	68.4	69.0	69.4	68.5	65.7	64.4	65.1	66.0	67.3	68.7	69.0	(24)	
	10%	DB	82.0	82.5	82.6	82.1	82.0	82.3	81.8	82.3	82.3	82.3	80.7	80.3	(25)	
		MCWB	69.1	69.2	69.6	69.6	68.6	66.3	64.7	65.7	66.7	68.2	69.0	69.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	73.2	73.8	73.4	72.0	71.9	71.3	72.0	72.0	72.8	72.4	72.2	(27)	
		MCDB	79.9	80.1	80.8	80.4	79.6	79.4	78.6	78.6	78.7	79.1	79.1	79.0	(28)	
	2%	WB	71.9	71.9	72.5	72.0	71.1	70.4	69.8	70.4	71.1	71.7	71.4	71.3	(29)	
		MCDB	79.3	79.6	80.1	79.8	79.1	78.6	77.4	77.6	78.0	78.4	78.3	78.5	(30)	
	5%	WB	71.1	71.2	71.6	71.3	70.5	69.5	68.7	69.6	70.2	70.9	70.7	70.6	(31)	
		MCDB	78.8	79.2	79.5	79.2	78.6	77.8	76.7	76.8	77.2	77.7	77.6	77.8	(32)	
	10%	WB	70.2	70.5	70.8	70.6	69.9	68.7	67.5	68.8	69.6	70.2	70.1	70.0	(33)	
		MCDB	78.3	78.6	78.8	78.3	78.1	76.9	76.2	76.2	76.6	77.0	77.0	77.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	15.5	15.4	14.8	15.9	18.3	19.7	18.5	16.6	15.3	13.6	13.1	(35)	
		MCDBR	18.0	18.9	18.8	17.7	17.5	20.6	21.1	20.8	19.3	18.5	16.7	16.4	(36)	
	5% WB	MCWBR	7.4	7.2	7.9	7.2	7.5	9.4	10.2	11.0	9.2	7.9	6.4	6.3	(37)	
		MCWBR	15.7	15.9	16.3	15.3	15.8	17.4	18.8	18.3	16.6	15.2	14.3	14.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.414	0.426	0.407	0.359	0.357	0.386	0.404	0.416	0.397	0.389	0.376	0.400	(40)	
		taud	2.376	2.343	2.426	2.584	2.567	2.430	2.330	2.305	2.376	2.409	2.488	2.422	(41)	
	Ebn at Noon		292	289	291	298	291	278	274	277	290	296	301	295	(42)	
		Edh at Noon	41	42	39	32	31	35	39	42	40	39	36	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		1720	1885	1853	1790	1765	1786	1836	1820	1817	1788	1690	1667	(44)	
	RadStd		167	96	111	141	96	102	82	109	115	122	125	136	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page